

Supplementary Materials: Calibrating Discrete Entropy Prediction Floors for Urban Crime Counts: A Multicity Evaluation

Minling Dang ¹, Yan Liu ¹, Ying Zhang ¹, Bin Guo¹ and Zhiwen Yu ^{2,1*}

S1. Estimator Sensitivity and Training-Period-Frozen Decision Populations

The estimator-sensitivity analysis evaluated the plug-in, Miller–Madow, and coherent observed-support Jeffreys–Dirichlet estimators across 16 prespecified specifications in each city. All 16 DELB-reduction point estimates were positive for every city and estimator (Table S1). The Jeffreys calculation places a symmetric posterior on observed joint states and coherently aggregates its marginals; it does not reconstruct unobserved support.

The estimator-specific randomization analysis applied all three estimators to the same 999 temporal-block, spatial-series, and circular-shift surrogates. Table S2 reports both Benjamini–Hochberg adjustment within each estimator’s nine city–design comparisons and global adjustment over all 27 comparisons. Figure S1 shows observed CMI relative to the estimator-specific null mean.

Table S1. Sparse-entropy estimator sensitivity. Positive specifications are counted among the 16 prespecified designs for each city–estimator pair.

City	Estimator	Positive specs	Primary CMI (bits)	Primary ΔL
Washington, DC	Jeffreys–Dirichlet	16/16	0.0690	0.0187
Washington, DC	Miller–Madow	16/16	0.0432	0.0123
Washington, DC	Plug-in	16/16	0.0526	0.0141
New York City	Jeffreys–Dirichlet	16/16	0.1603	0.0624
New York City	Miller–Madow	16/16	0.1314	0.0604
New York City	Plug-in	16/16	0.1394	0.0550
Vancouver	Jeffreys–Dirichlet	16/16	0.1904	0.0940
Vancouver	Miller–Madow	16/16	0.1423	0.0919
Vancouver	Plug-in	16/16	0.1597	0.0844

Table S2. City-level estimator-specific randomization calibration. Ranges cover nine city–reference-design comparisons per estimator; the final column gives rejection counts under the within-estimator nine-test and global 27-test corrections.

Estimator	Observed - null mean	Raw p	Min. q_9	Min. q_{27}	Rejections
Jeffreys–Dirichlet	[-0.0536, -0.0016]	[0.984, 1.000]	1.000	1.000	0/9; 0/27
Miller–Madow	[-0.0423, -0.0012]	[0.903, 1.000]	1.000	1.000	0/9; 0/27
Plug-in	[-0.0470, -0.0012]	[0.980, 1.000]	1.000	1.000	0/9; 0/27

The decision-population comparison contrasts the training-period-frozen coverage-conditioned estimand with the complete crime-supported deployment-population estimand. Both populations were fixed before validation and testing; this is therefore an estimand comparison, not a post-outcome spatial selection. Table S3 and Figure S2 show that expanding the spatial population changes the magnitude of CMI and DELB reduction, especially in New York City and Vancouver. These differences are descriptive and do not identify a bicycle effect.

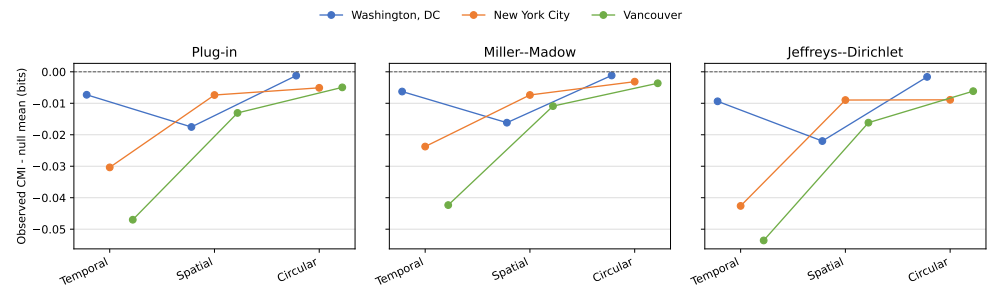


Figure S1. Estimator sensitivity of city-level randomization calibration. Each point is the observed CMI minus the mean of the same design-conditioned reference distribution. Nonzero null centers are allowed.

Table S3. Training-period-frozen analysis-domain sensitivity. Both spatial populations were defined before validation and testing.

City	Analysis domain	Grids	N	CMI	L^0	L^B	ΔL	Relative
Washington, DC	Frozen bicycle-covered domain	181	198195	0.0432	0.1567	0.1443	0.0123	7.9%
Washington, DC	Complete crime-supported domain	186	203670	0.0420	0.1535	0.1416	0.0119	7.7%
New York City	Frozen bicycle-covered domain	198	216810	0.1314	0.3503	0.2898	0.0604	17.3%
New York City	Complete crime-supported domain	806	882570	0.0348	0.1382	0.1288	0.0094	6.8%
Vancouver	Frozen bicycle-covered domain	34	37230	0.1423	0.5113	0.4194	0.0919	18.0%
Vancouver	Complete crime-supported domain	141	154395	0.0377	0.1774	0.1659	0.0115	6.5%

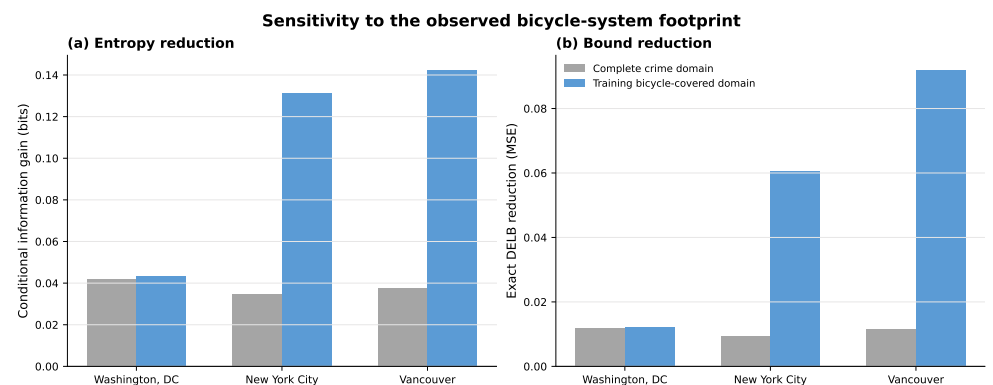


Figure S2. Training-period-frozen decision-population comparison. The complete crime-supported deployment population and the coverage-conditioned bicycle-covered population define different estimands.

Table S4. Pooled city-level conditional entropy and DELB results in the training bicycle-covered domain. Brackets report centered 95% seven-day block-bootstrap intervals for CMI.

City	N	H^0	H^B	CMI [95% CI]	L^0	L^B	ΔL	R^L
Washington, DC	198,195	0.783	0.740	0.043 [0.040, 0.047]	0.157	0.144	0.012	7.9%
New York City	216,810	1.293	1.162	0.131 [0.123, 0.140]	0.350	0.290	0.060	17.3%
Vancouver	37,230	1.563	1.421	0.142 [0.126, 0.159]	0.511	0.419	0.092	18.0%

Table S5. City-level DELB point estimates for two training-period-frozen analysis populations. The deployment population uses the complete crime-supported domain; the bicycle-covered population uses its bicycle-covered subset.

City	Population	Grids	<i>N</i>	CMI (bit)	L^0	L^B	ΔL
Washington, DC	Deployment (complete)	186	203,670	0.0420	0.1535	0.1416	0.0119
Washington, DC	Coverage-conditioned	181	198,195	0.0432	0.1567	0.1443	0.0123
New York City	Deployment (complete)	806	882,570	0.0348	0.1382	0.1288	0.0094
New York City	Coverage-conditioned	198	216,810	0.1314	0.3503	0.2898	0.0604
Vancouver	Deployment (complete)	141	154,395	0.0377	0.1774	0.1659	0.0115
Vancouver	Coverage-conditioned	34	37,230	0.1423	0.5113	0.4194	0.0919

S2. Local Estimates and Lag-Specificity Details

Table S6. Spatially localized information value. The final column gives global Moran’s *I* for DELB reduction with two-sided permutation *p*-values in parentheses.

City	Eligible/Covered	Median CMI	Median ΔL	Median R^L	Positive screen	Moran’s <i>I</i> (<i>p</i>)
Washington, DC	146/181	0.042	0.013	9.6%	137/146	0.164 (0.002)
New York City	160/198	0.093	0.050	14.3%	155/160	0.038 (0.261)
Vancouver	33/34	0.138	0.094	17.4%	33/33	0.171 (0.022)

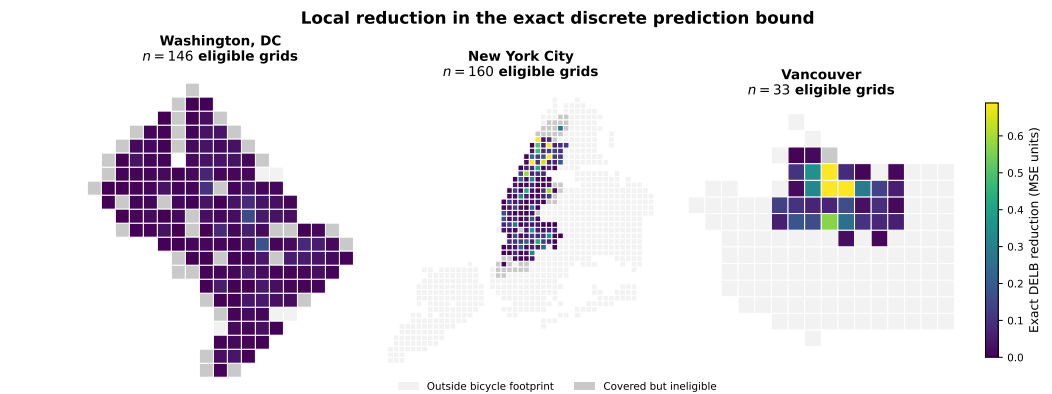


Figure S3. Local DELB-reduction point estimates for eligible cells. None passed all three randomization references.

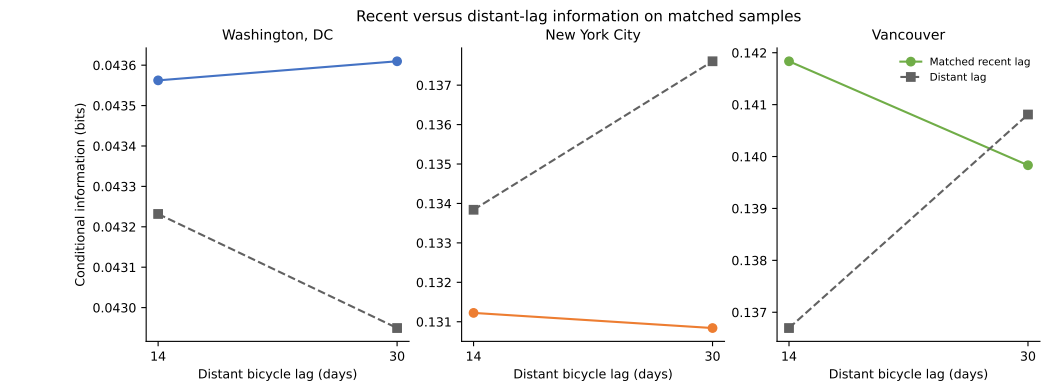


Figure S4. Matched recent- and distant-lag CMI estimates. Their similarity shows that the positive primary point estimate is not specific to the one-day mobility horizon.

S3. Sample-Size and Multiscale Details

S3.1. Nested-domain sampling and normalization

The comparison domain comprised complete 2 km parents, their four 1 km children, and all sixteen 500 m descendants. Crime and bicycle totals reconciled across this nested footprint. Weekly records summed seven complete days; a one-period lag referred to the preceding aggregation period. Sample fractions retained all grids within selected blocks. Daily panels used seven-day blocks and weekly panels four-week blocks, stratified by year and season. Fractions were nested within each of 50 frozen repetitions; a five-point expanding-window design separately diagnosed chronological accumulation and distribution drift.

Estimator stability for an entropy, CMI, or DELB estimate was

$$S_N(\hat{\theta}) = \frac{|\hat{\theta}_N - \hat{\theta}_{N_{\max}}|}{\max(|\hat{\theta}_{N_{\max}}|, 10^{-12})}. \quad (1)$$

This is deviation from the full-sample estimate, not error relative to unknown population truth. For cell area A in km^2 and duration τ in days, the deterministically scaled intensity target has

$$L_{\text{int,MSE}}^j(A, \tau) = \frac{L^j(A, \tau)}{(A\tau)^2}, \quad L_{\text{int,RMSE}}^j(A, \tau) = \frac{\sqrt{L^j(A, \tau)}}{A\tau}, \quad j \in \{0, B\}. \quad (2)$$

Temporal blocks in the randomization analysis contained seven days or four weeks; circular shifts were separated by at least 31 days or five weeks. At fractions below 100%, the lowest-index frozen thinning path entered randomization calibration, while stability summarized all 50 paths. Miller–Madow CMI, DELB reduction, occupied support, singleton share, and effective degrees of freedom were recomputed after every transformation.

S3.2. Truth-referenced semi-synthetic protocol

Six flow-quantile-spanning grids were selected deterministically in each city. Observed baseline and bicycle states were held fixed; current crime counts were regenerated from a negative-binomial model. Conditional means used 20-observation shrinkage. Raw method-of-moments dispersion estimates were retained as diagnostics, but simulation dispersion was bounded below by 2.0 because the finer-scale supports could not attain 0.12-bit CMI under the bounded multiplicative mean-shift family at their raw dispersion. The bicycle-state coefficient was calibrated by numerical enumeration so that $H(C | \mathcal{S}^0) - H(C | \mathcal{S}^0, B)$ equaled the target CMI. Both conditional entropies were mapped through the same inverse integer-lattice envelope to obtain population L^0 , L^B , and ΔL . All three estimators were evaluated on the same generated datasets. Truth-referenced outcomes were bias, RMSE, relative RMSE for positive truth, Monte Carlo standard deviation, and sign recovery. Power used Miller–Madow CMI and exact baseline-state-matched permutation without replacement among observations having identical baseline-state codes. The 12 high-resolution cells crossed $N \in \{150, 365\}$, $\text{CMI} \in \{0.03, 0.06\}$, and three scales. Plus-one upper-tail p -values used a 0.05 threshold. The 80% minimum detectable CMI was the smallest tested positive CMI reaching 80% power; linear interpolation between adjacent levels was reported as a grid-based approximation, not an additional simulated effect level.

The scale–sample support audit shows how the fixed design changes with generated record length. Figure S9 reports occupied baseline and baseline–bicycle states, observations per occupied joint state, and the share movable under exact baseline-state matching.

Table S7. Median relative deviation from the 100% estimate across the 18 city–space–time cells. Quantiles over thinning repetitions are design-variability summaries, not population confidence intervals.

Sample fraction	Baseline DELB	Bicycle DELB	DELB reduction	CMI
20%	0.631	0.646	0.630	0.271
40%	0.423	0.448	0.341	0.250
60%	0.231	0.249	0.179	0.172
80%	0.101	0.114	0.070	0.087
100%	0.000	0.000	0.000	0.000

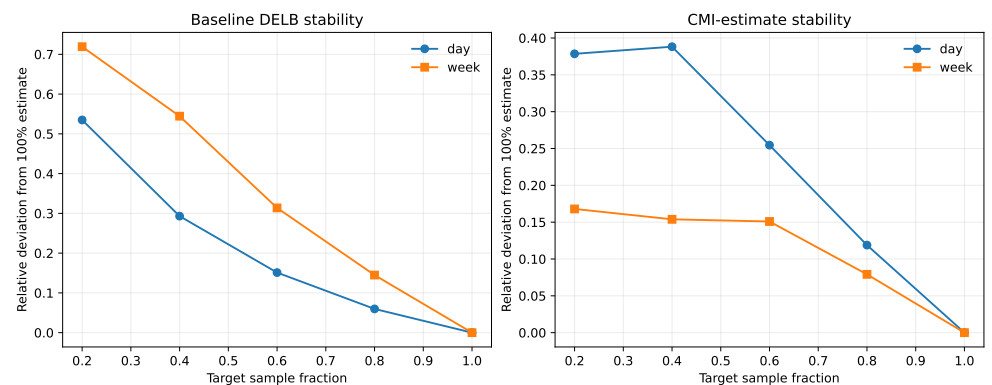


Figure S5. Relative deviation from the full-sample baseline DELB and CMI estimate under nested block thinning. The curves measure estimator stability, not distance from population truth.

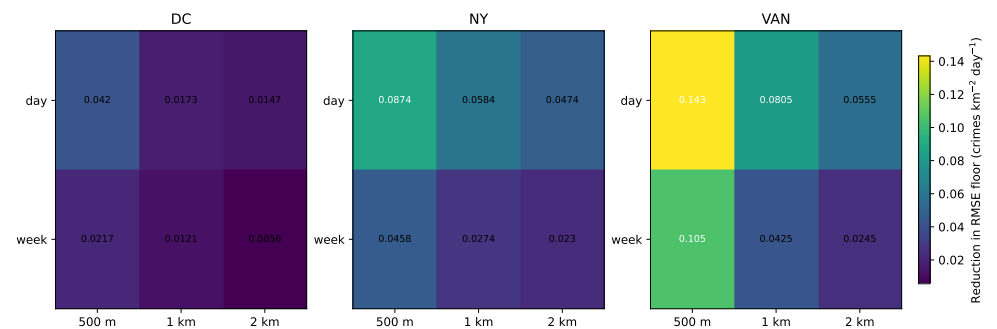


Figure S6. Full-sample Miller–Madow reduction in the deterministically scaled intensity-RMSE floor by city, spatial resolution, and temporal resolution.

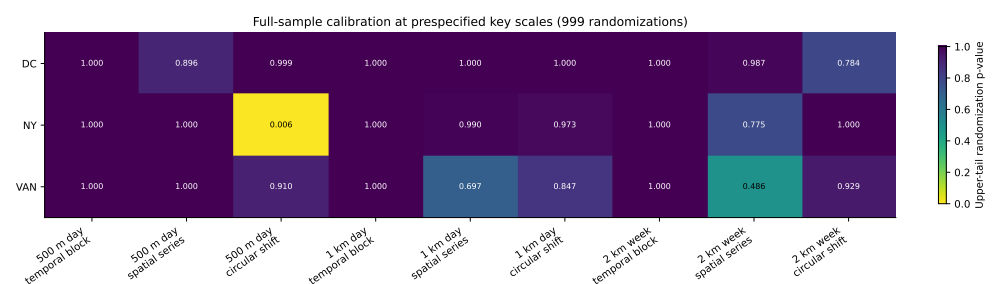


Figure S7. Full-sample upper-tail randomization p -values at the 500 m–day, 1 km–day, and 2 km–week key scales.

Table S8. Multiscale upper-tail randomization calibration. Discoveries use a Benjamini–Hochberg adjustment across all 270 tests.

Null design	Tests	Min. p	Median p	Obs. > null	FDR discoveries
Temporal block	90	0.873	1.000	0	0
Spatial series	90	0.130	0.945	12	0
Circular shift	90	0.005	0.917	21	0

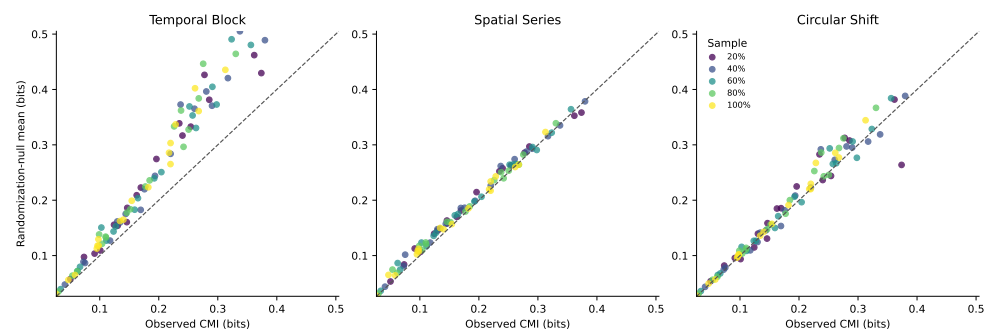


Figure S8. Observed Miller–Madow CMI versus the randomization-reference mean across 270 city–space–time–sample–design comparisons. The dashed line is equality; a nonzero reference center is allowed and is not population CMI.

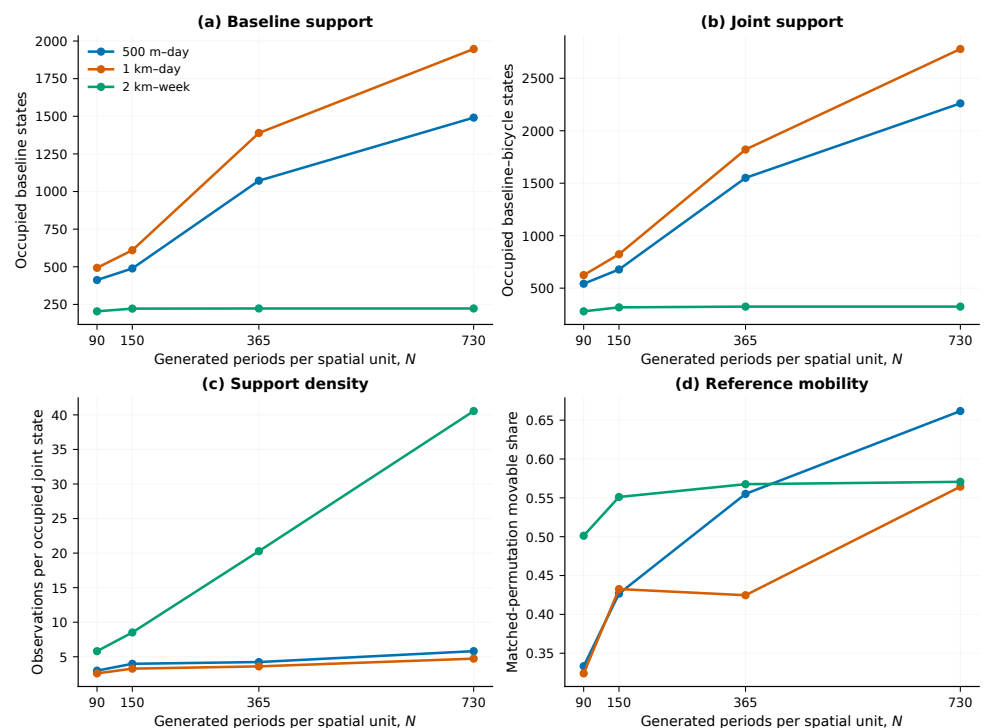


Figure S9. Support complexity of the semi-synthetic scale–sample designs. For 2 km–week, values beyond the observed record length use the prespecified within-grid state-sequence replay.

Figure S10 gives the truth-referenced error surface by estimator, scale, and generated sample length. Color limits are shared within each metric row, so cells are comparable across estimators.

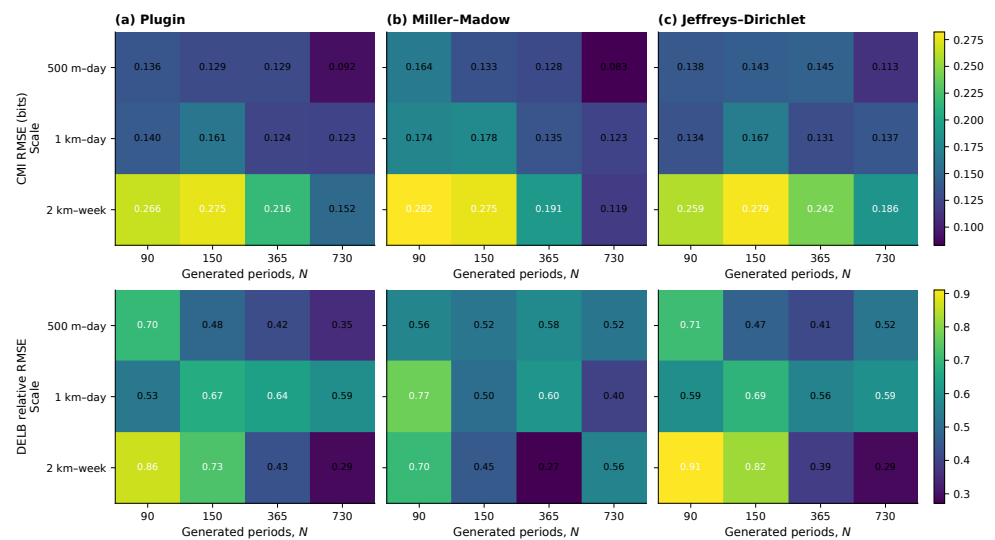


Figure S10. Median truth-referenced CMI RMSE and DELB relative RMSE over the three positive population-CMI levels.

S4. Crime-Type Heterogeneity

Table S9. Pooled crime-type heterogeneity in bicycle conditional information and DELB reduction. Intervals are centered 95% seven-day block-bootstrap intervals for CMI.

City	Crime type	CMI	95% CI	ΔL	R^L
Washington, DC	Property theft	0.035	[0.032, 0.038]	0.009	7.1%
Washington, DC	Vehicle theft	0.012	[0.011, 0.014]	0.002	6.7%
Washington, DC	Burglary	0.006	[0.005, 0.007]	0.001	7.7%
New York City	Property theft	0.113	[0.105, 0.121]	0.045	15.8%
New York City	Vehicle theft	0.023	[0.021, 0.025]	0.004	10.1%
New York City	Burglary	0.028	[0.026, 0.031]	0.006	9.3%
Vancouver	Property theft	0.114	[0.100, 0.128]	0.060	14.9%
Vancouver	Vehicle theft	0.013	[0.011, 0.015]	0.002	8.1%
Vancouver	Burglary	0.041	[0.034, 0.048]	0.010	9.1%

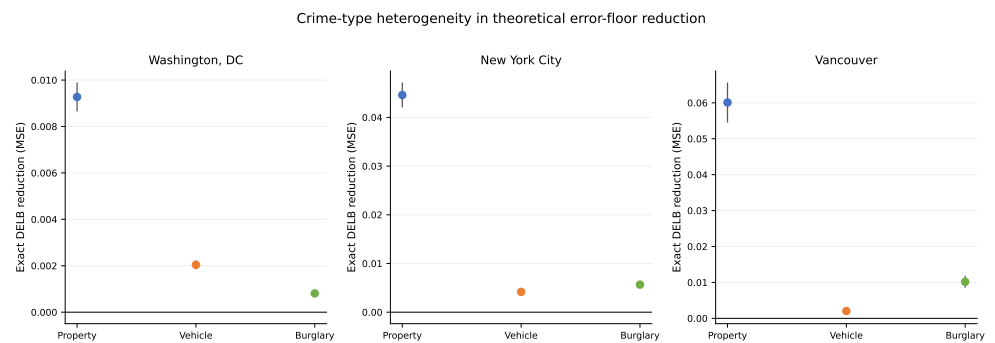


Figure S11. DELB-reduction point estimates by harmonized crime category and city. Intervals measure block-bootstrap sampling variation and are not category-specific randomization tests.

S5. Alternative Specifications and Rolling Origins

Table S10 reports the additional matched-reference sensitivities. The lower-cardinality specification replaces grid identity with training-only crime-intensity quintiles. Calendar-month and positive-flow-quintile specifications intentionally increase temporal or bicycle-state resolution; neither exceeded its matched-reference center.

The bicycle-state mapping changes the magnitude of the uncalibrated Miller-Madow CMI point estimate. For zero plus a positive-flow median split, the Washington, DC, New York City, and Vancouver estimates were 0.02934, 0.09247, and 0.08310 bits. The primary zero-plus-tertiles mapping gave 0.04316, 0.13137, and 0.14233 bits, while zero plus positive-flow quintiles gave 0.07169, 0.19650, and 0.24360 bits. These values come from the frozen alternative-mapping results; higher bin resolution alone is not calibrated evidence of additional source information. The finest mapping's matched-reference comparison is reported in Table S10.

Table S10. Additional robustness specifications. Differences compare observed Miller-Madow CMI with the mean baseline-state-matched reference.

Specification	City	CMI	Null mean	Difference	<i>p</i>
Lower-cardinality	Washington, DC	0.01315	0.01291	0.00024	0.165
Lower-cardinality	New York City	0.03625	0.03520	0.00105	0.001
Lower-cardinality	Vancouver	0.07186	0.07178	0.00008	0.480
Month replaces season	Washington, DC	0.07154	0.07441	-0.00287	1.000
Month replaces season	New York City	0.15531	0.17154	-0.01623	1.000
Month replaces season	Vancouver	0.19312	0.21794	-0.02482	1.000
Positive-flow quintiles	Washington, DC	0.07169	0.07380	-0.00211	1.000
Positive-flow quintiles	New York City	0.19650	0.19922	-0.00272	0.995
Positive-flow quintiles	Vancouver	0.24360	0.25224	-0.00864	1.000

Table S11. Reconciliation of city-level ΔL values. Every row uses Miller-Madow conditional entropy and the exact inverse lattice envelope. The population, period, and analytical purpose are explicit for each estimate; values are in MSE units.

City	Population	Period	Purpose	ΔL
Washington, DC	Bicycle-covered	2020–2022	Primary information analysis	0.01232
Washington, DC	Complete crime-supported	2020–2022	Deployment-population comparison	0.01188
Washington, DC	Bicycle-covered	Jul–Dec 2022	Matched held-out MSE comparison	0.01511
New York City	Bicycle-covered	2020–2022	Primary information analysis	0.06044
New York City	Complete crime-supported	2020–2022	Deployment-population comparison	0.00940
New York City	Bicycle-covered	Jul–Dec 2022	Matched held-out MSE comparison	0.03363
Vancouver	Bicycle-covered	2020–2022	Primary information analysis	0.09191
Vancouver	Complete crime-supported	2020–2022	Deployment-population comparison	0.01150
Vancouver	Bicycle-covered	Jul–Dec 2022	Matched held-out MSE comparison	0.05095

Annual bicycle-covered CMI estimates for 2020, 2021, and 2022 were 0.0657, 0.0651, and 0.0599 bits in Washington, DC. The corresponding values were 0.0561, 0.1162, and

Table S12. Coverage dilution and common-top-code sensitivity. The top-code of 9 was fixed as the ceiling of the pooled training-domain 99.5th percentile.

City	Covered share	Covered CMI	Implied pooled	Reported pooled	Top-coded CMI
Washington, DC	0.973	0.04316	0.04200	0.04200	0.04316
New York City	0.246	0.13137	0.03227	0.03478	0.13067
Vancouver	0.241	0.14233	0.03432	0.03770	0.13917

Outside-footprint CMI was 0.00000, 0.00333, and 0.00445 bits for Washington, DC, New York City, and Vancouver, respectively. Its observation-share-weighted contribution to complete-domain CMI was 0.00000, 0.00251, and 0.00338 bits. Because grid identity belongs to the conditioning state, these terms reconcile the reported pooled values up to rounding.

Table S13. Empirical two-slack decomposition for all four model families and both information sets. I^0/I^B denotes residual error–state dependence and KL^0/KL^B denotes marginal error-shape mismatch, in bits, for baseline/bicycle-aware predictors. The absolute identity residual was below 3.31×10^{-13} bits in every cell.

City	Model	I^0	I^B	KL^0	KL^B
Washington, DC	State mean	0.769	0.828	0.275	0.276
Washington, DC	Poisson GLM	0.784	0.862	0.250	0.243
Washington, DC	Negative-binomial GLM	0.795	0.859	0.239	0.236
Washington, DC	Poisson gradient boosting	0.799	0.865	0.238	0.234
New York City	State mean	1.476	1.555	0.197	0.186
New York City	Poisson GLM	1.445	1.544	0.161	0.156
New York City	Negative-binomial GLM	1.436	1.530	0.171	0.172
New York City	Poisson gradient boosting	1.413	1.472	0.192	0.188
Vancouver	State mean	1.258	1.378	0.251	0.252
Vancouver	Poisson GLM	1.269	1.395	0.224	0.224
Vancouver	Negative-binomial GLM	1.266	1.393	0.221	0.223
Vancouver	Poisson gradient boosting	1.272	1.395	0.237	0.242

0.1064 in New York City and 0.2236, 0.1717, and 0.1690 in Vancouver. These point estimates show temporal variation over the pandemic-era study window and are not substitutes for the matched-reference calibration.

Table S14. Robustness summary across information specifications and rolling forecasts. Positive theory specifications count Miller–Madow DELB reductions across 16 pre-specified alternatives. Rolling forecasts comprise 12 monthly origins for each of four model families. Positive Δ MSE favors the bicycle-aware model.

City	Positive theory	Positive origins	State mean		Gradient boosting	
			Positive	Mean Δ MSE	Positive	Mean Δ MSE
Washington, DC	16/16	23/48	66.7%	0.001	41.7%	0.000
New York City	16/16	26/48	91.7%	0.030	100.0%	0.076
Vancouver	16/16	20/48	66.7%	0.006	41.7%	-0.006

S6. Additional Descriptive and Finite-Sample Figures

Figures S13 and S14 are descriptive audits. They precede conditioning and are not evidence of CMI, a DELB reduction, or a causal mobility effect.

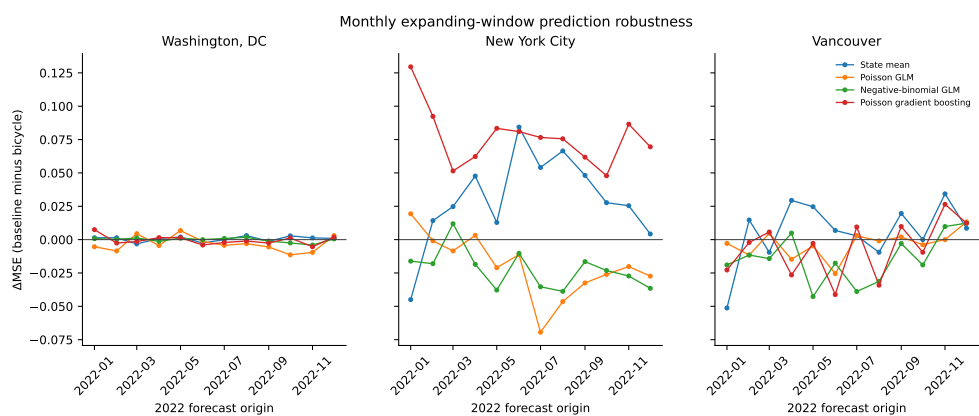


Figure S12. Expanding-window monthly prediction results. Positive ΔMSE favors the bicycle-aware model; stability remains model-specific.

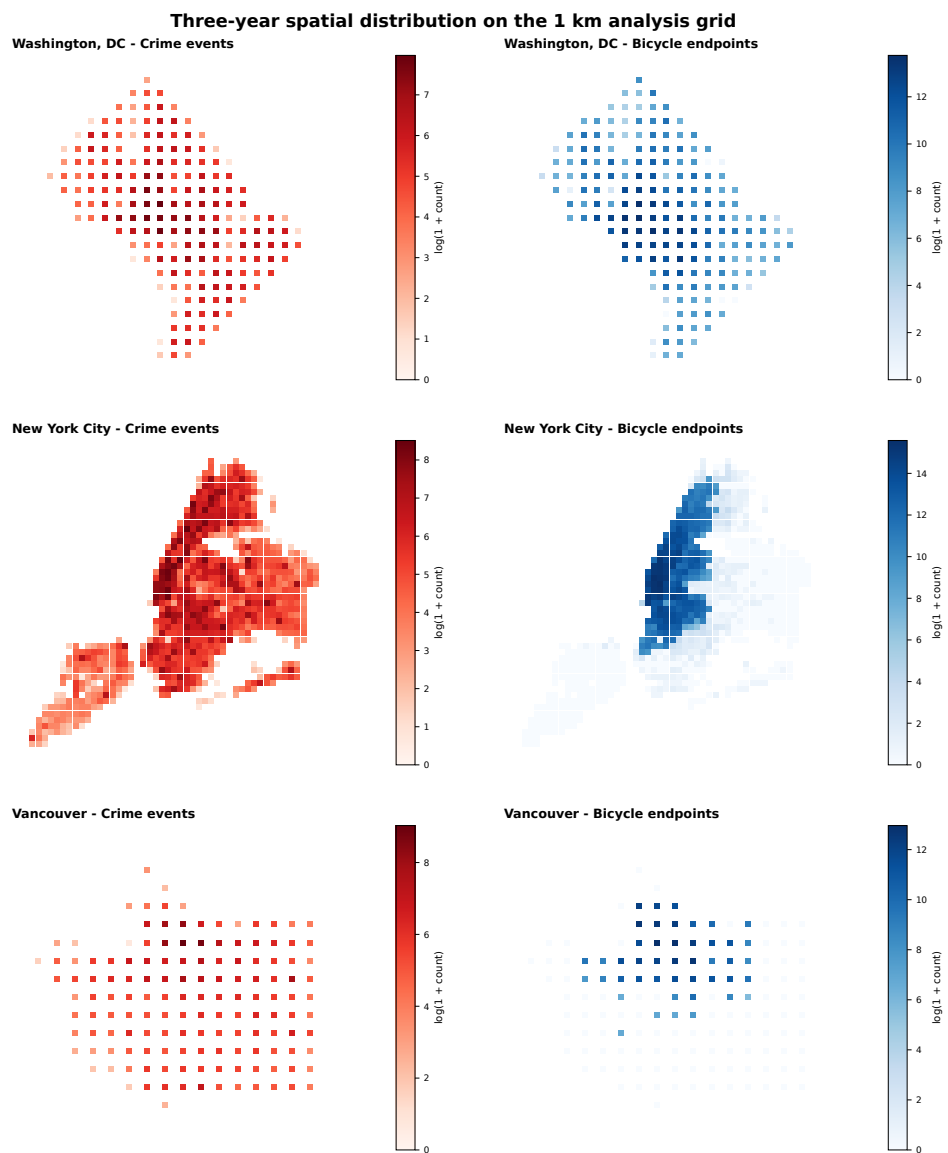


Figure S13. Spatial concentration of harmonized property-crime counts and bicycle endpoint flow, 2020–2022.

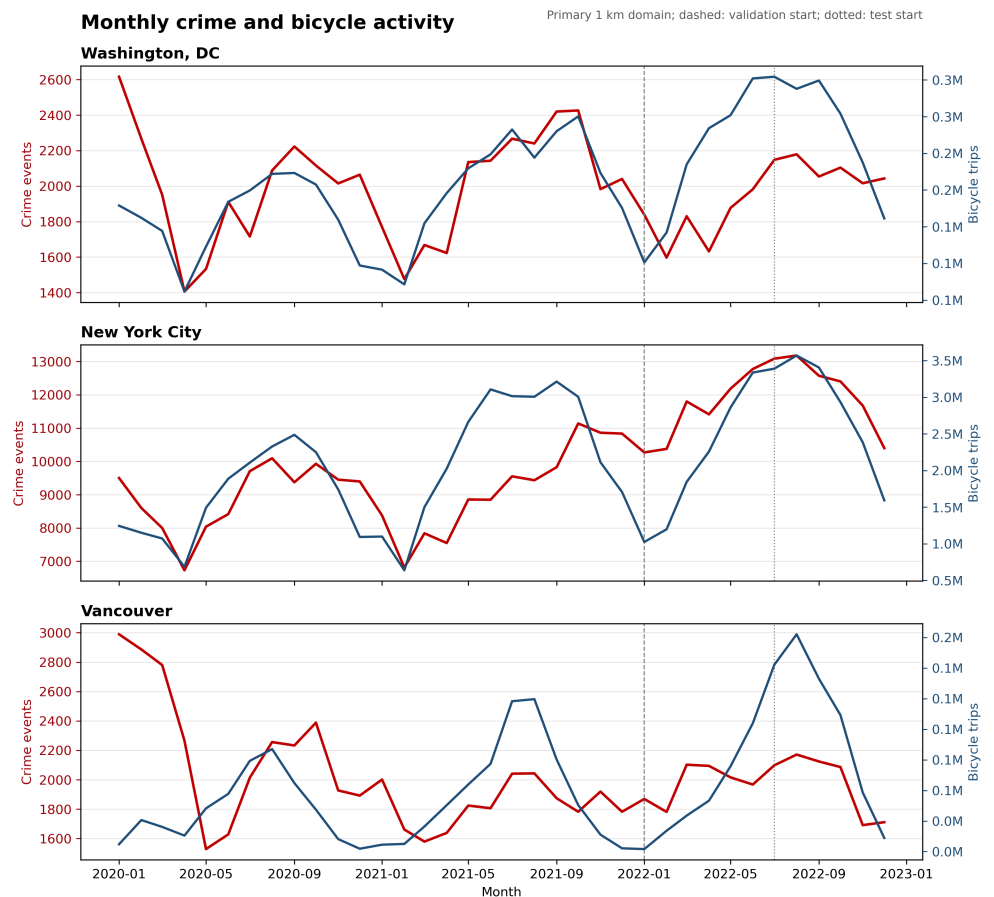


Figure S14. Monthly crime counts and bicycle endpoint flow by city, 2020–2022.

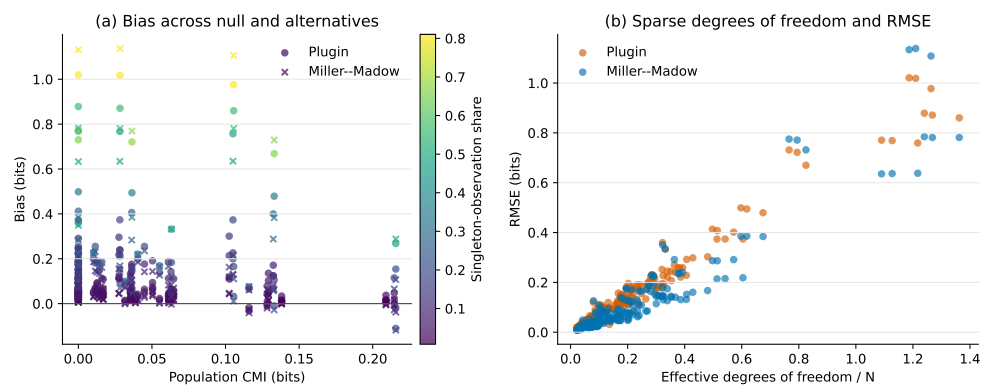


Figure S15. Estimator bias and RMSE over the complete sparse-state simulation grid.

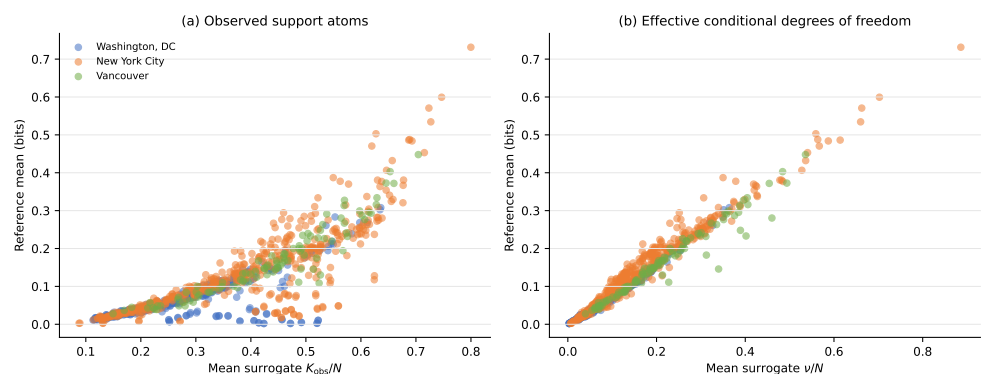


Figure S16. Supplementary support-metric relationships for simulation and empirical null summaries.

Table S15. Association of support diagnostics with CMI estimator behavior. Entries are Spearman correlations. The first two columns use population-null simulations; the last two use 1,017 city-grid-null summaries.

Metric	Plug-in bias	MM bias	Null mean	Observed-null
K_{obs}/N	0.840	0.891	0.835	-0.653
S_1	0.819	0.909	0.866	-0.656
ν/N	0.989	0.950	0.988	-0.791

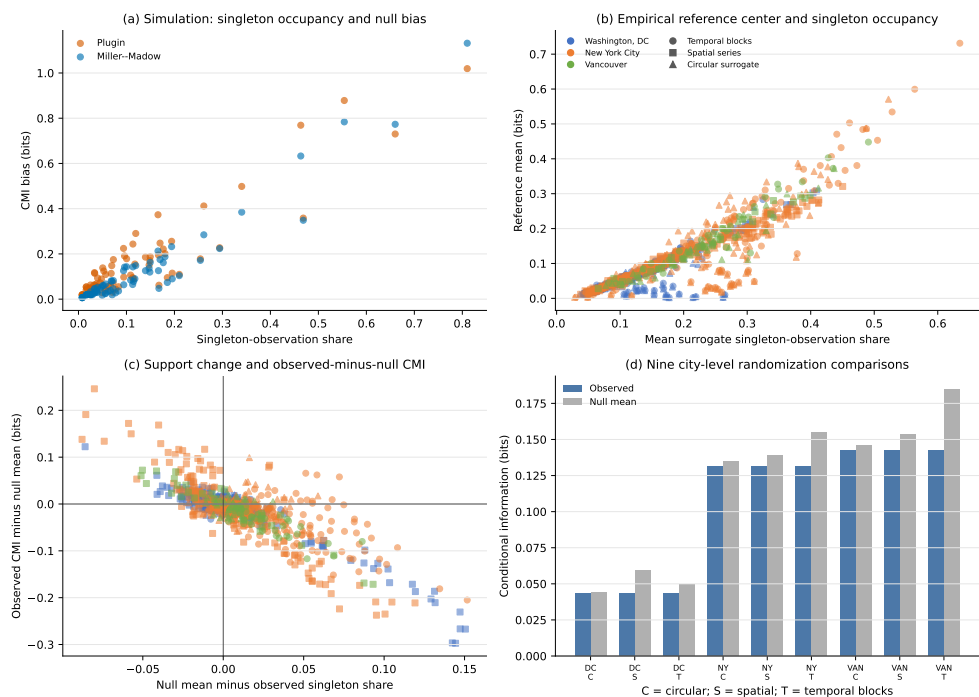


Figure S17. Support structure and randomization-reference centering. Simulation panels use known zero population CMI; empirical panels use surrogate-specific support recomputed after every transformation. Lines are descriptive associations.

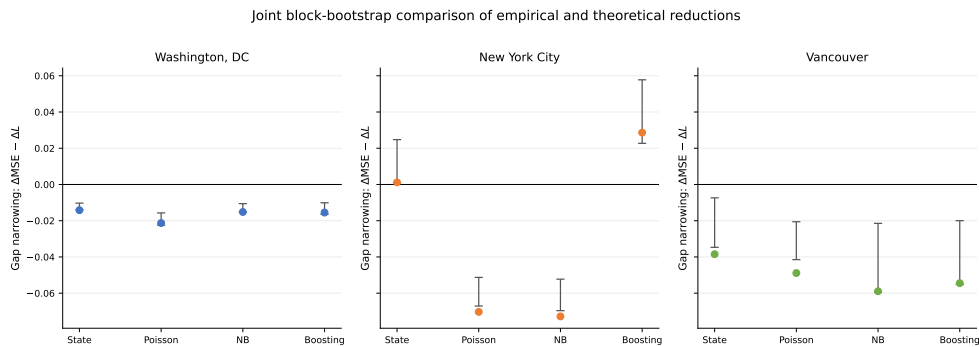


Figure S18. Predictability-gap narrowing after adding bicycle state. Positive values mean empirical MSE fell by more than the estimated DELB; negative values indicate a larger unused-information gap.

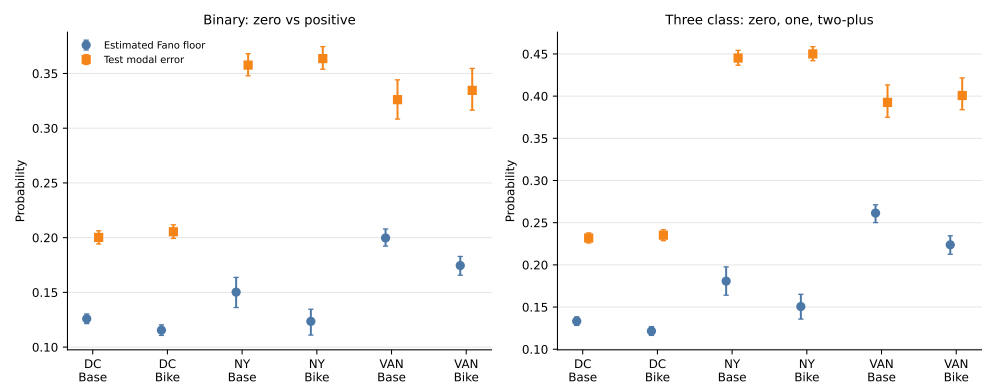


Figure S19. Centered seven-day block-bootstrap intervals for estimated Fano floors and held-out modal-classifier errors.

Table S16. Miller–Madow Fano floors and held-out statewise modal-classifier errors. The interval refers to the baseline-minus-bicycle floor reduction. The final column gives baseline/bicycle-aware counts of test observations assigned through the pretest global-mode fallback because their states were unseen before testing.

City	Target	Base floor	Bike floor	Reduction [95% CI]	Test error	Unseen test observations
Washington, DC	Binary occurrence	0.126	0.115	0.010 [0.010, 0.011]	0.200/0.205	865/2,071
Washington, DC	Three-level count	0.133	0.122	0.012 [0.011, 0.013]	0.232/0.235	865/2,071
New York City	Binary occurrence	0.150	0.124	0.027 [0.024, 0.029]	0.358/0.364	2,058/4,408
New York City	Three-level count	0.181	0.151	0.030 [0.027, 0.033]	0.445/0.450	2,058/4,408
Vancouver	Binary occurrence	0.200	0.175	0.025 [0.022, 0.028]	0.326/0.335	234/812
Vancouver	Three-level count	0.261	0.224	0.038 [0.034, 0.041]	0.393/0.401	234/812

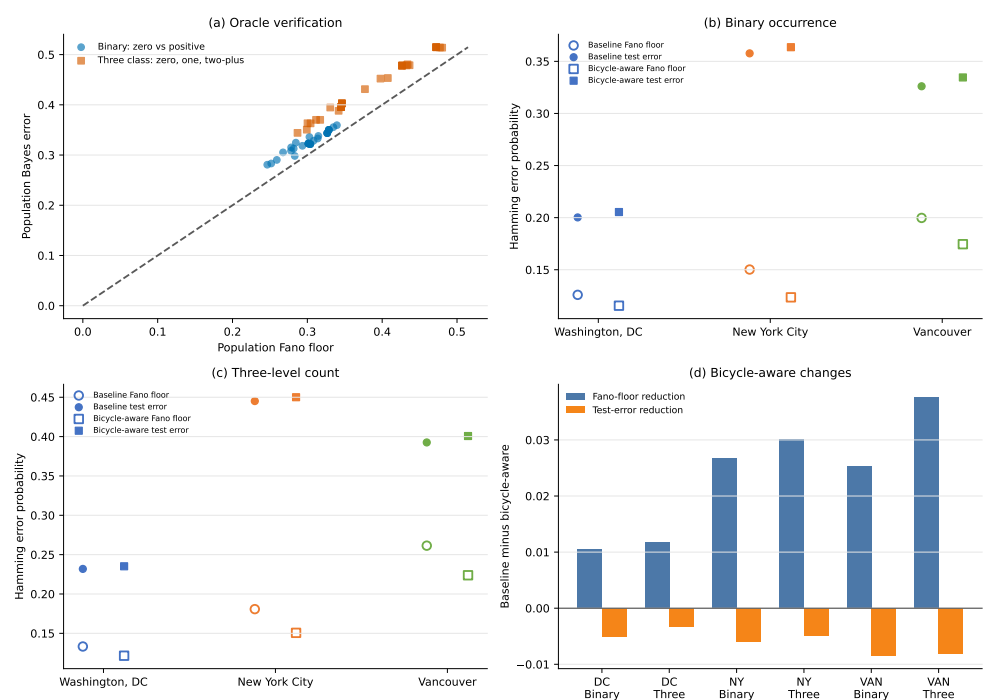


Figure S20. Fano classification-loss comparison. Oracle panels compare true Bayes errors with true Fano floors; empirical panels compare estimated pretest floors with held-out modal-classifier errors.

S7. Empirical Harmonization and Audit Details

S7.1. Crime-Category Crosswalk

Table S17 records the source-offense mapping. The standardized field is `crime_type_unified`. No category was remapped after harmonization. New York City's vehicle-theft category includes *Petit Larceny of Motor Vehicle* and, when present in the source data, *Grand Larceny of Motor Vehicle*.

Table S17. Source offenses represented by the three harmonized property-crime categories.

Unified category	Vancouver	New York City	Washington, DC
Property theft	Other Theft; Theft from Vehicle; Theft of Bicycle	Petit Larceny	Theft-Other; Theft F/Auto
Vehicle theft	Theft of Vehicle	Petit Larceny of Motor Vehicle; Grand Larceny of Motor Vehicle, if present	Motor Vehicle Theft
Burglary	Break and Enter Commercial; Break and Enter Residential/Other	Burglary	Burglary

S7.2. Endpoint-Coordinate Provenance

Table S18. Mutually exclusive coordinate outcomes among retained bicycle-trip endpoints. "Reference" denotes deterministic station-key matching, not statistical or outcome-based imputation. Percentages use all retained start and end endpoints within each city as the denominator.

City	Endpoints	Direct source	Reference	Unresolved	Non-public
Washington, DC	16,887,154	93.532895%	6.361190%	0.105915%	0.000000%
New York City	153,018,686	99.906369%	0.000006%	0.093594%	0.000031%
Vancouver	4,511,360	0.000000%	95.513437%	4.322067%	0.164496%

The table is generated by `empirical/reviewer4_coordinate_audit.py` from the frozen endpoint-coordinate provenance data. Direct source, reference-resolved, unresolved, and non-public counts sum exactly to the retained endpoint total in every city.

The coordinate-source sensitivity holds the 1 km crime panel, time split, and frozen bicycle-covered grids fixed. At each endpoint, it subtracts flow only when the recorded coordinates came from deterministic station-reference matching; a directly located opposite endpoint remains eligible. Positive flow thresholds are refitted on the unchanged training period, as in the primary analysis. The 199-replicate paired randomizations use new fixed seeds; the Washington, DC circular-shift comparison was also checked with 999 replicates. Values are post hoc diagnostics and are not added to the prespecified primary multiple-testing family.

Table S19. Effect of coordinate-source restriction on primary city-level information estimates. The fixed crime sample contains 198,195 observations in Washington, DC and 216,810 in New York City. CMI is the Miller–Madow point estimate; ΔL is the exact DELB reduction in MSE units. “Removed flow” is the share of all endpoint contributions in the crime-supported primary grid domain during 2020–2022.

City	Endpoint rule	Removed flow	CMI (bits)	ΔL (MSE)
Washington, DC	All eligible	0.0000%	0.043157	0.012324
Washington, DC	Direct source only	6.3929%	0.063098	0.017857
New York City	All eligible	0.0000%	0.131369	0.060444
New York City	Direct source only	< 0.000001%	0.131369	0.060444

Table S20. Post hoc coordinate-source randomization sensitivity. Upper-tail p -values use 199 surrogates per design; the Washington, DC circular-shift comparison is separately confirmed with 999 surrogates. The 999-replicate check uses the same fixed sensitivity seed for the paired endpoint rules.

City	Endpoint rule	Temporal	Spatial	Circular	Circular, 999
Washington, DC	All eligible	1.000	1.000	0.955	0.948
Washington, DC	Direct source only	1.000	1.000	0.005	0.007
New York City	All eligible	1.000	1.000	0.995	–
New York City	Direct source only	1.000	1.000	0.995	–

The Washington, DC point estimates are sensitive to excluding reference-located endpoints. The circular-shift result changes direction, whereas temporal and spatial randomizations remain non-separating. This contrast is exploratory because the restriction was devised after review, and the three references test different temporal and spatial structures. The New York City binned bicycle state and estimates are unchanged: only one in-domain flow contribution is removed. Vancouver has no source-only endpoint coordinates, so these tables do not establish three-city robustness. The sensitivity script and machine-readable tables are in `empirical/reviewer4_coordinate_sensitivity.py` and `empirical/runs/reviewer4_sensitivity_20260923/tables`.

S7.3. Reproducible Analysis Components

The reproducible analysis comprises data harmonization, leakage-safe panel construction, numerical verification of the integer entropy envelope, city-level and local DELB estimation, matched held-out prediction, crime-category and rolling-origin analyses, randomization calibration, known-truth simulation, support diagnostics, Fano comparisons, multiscale evaluation, estimator sensitivity, and baseline-state-matched power analysis. Version-controlled scripts preserve the configurations, random seeds, reason-coded exclusions, intermediate checks, and machine-readable outputs for each component.

S7.4. Robustness and Null-Calibration Scope

The robustness analysis contains 144 theory rows (16 specifications, three estimators, and three cities), 144 monthly rolling model–origin results, and three bootstrap block lengths per city. The randomization analysis contains 8,991 city-level reference replicates, 1,015,983 local null replicates, 339 joint local screens, and matched 14- and 30-day lag diagnostics. These artifacts are retained in their accepted run directories. The null designs are treated as finite-sample diagnostics rather than causal or exact conditional randomization tests. Every article figure was created by a versioned Python entry point and was not manually edited. Estimator sensitivity comprises 26,973 city-level reference statistics

Table S21. The four-part DELB theorem package and the distinct question answered by each component. The contribution is their joint organization for non-anticipating integer prediction; the underlying maximum-entropy and information-projection ingredients remain prior art.

Component	Mathematical object	Prediction question answered
Exact inverse-lattice floor	$\mathcal{H}_Z^{-1}[H(X_t \mid \mathcal{F}_{t-m})]$	What MSE floor is imposed by the target uncertainty and the integer lattice, independently of the forecasting algorithm?
Predictor-dependent refinement	$\mathcal{H}_Z^{-1}[H(X_t \mid \mathcal{F}_{t-m}) + I(\tilde{E}_t; \mathcal{F}_{t-m})]$	How is the lower bound strengthened for a particular predictor whose error still retains information about the available history?
Necessary and sufficient attainability conditions	$P_E = q_D$ and $I(E; \mathcal{F}) = 0$	When does a predictor attain the universal DELB, rather than merely satisfy the inequality?
Two-component prediction slack	$I(E; \mathcal{F}) + D_{\text{KL},2}(P_E \parallel q_D)$	Is nonattainment attributable to residual history dependence, error-shape mismatch, or both?

(8,991 per estimator), with identical Miller–Madow statistics in the primary and sensitivity analyses. The restricted-randomization analysis contains 2,997 empirical city-level replicates and 2,400 semi-synthetic calibration summaries. The scale–sample analysis retains 12 fixed-design panels and 9,600 generated datasets, together with truth, power, minimum-detectable-CMI, and support summaries.

Descriptive figures, estimator sensitivity, training-period-frozen decision-population comparison, and the extended local, multiscale, crime-type, rolling-origin, and Fano diagnostics are supplied in the independent Supplementary Materials file.

S8. Supplementary Theory and Proofs

This appendix places the DELB relative to established bounds, states the supplementary Fano counterpart, and proves all main-text theoretical results. All logarithms in entropy expressions are to base 2; $\ln$ denotes the natural logarithm.

S8.1. Scope Relative to Established Bounds

The DELB is a non-anticipating integer-prediction specialization and synthesis rather than a claim to a new primitive maximum-entropy law. Shannon’s source-coding and rate–distortion arguments, Bayes-risk inequalities, Ziv–Zakai bounds, and Fano-type error bounds address different operational objects [1,8–10,12]. Massey’s integer entropy–variance relation and later refinements provide the lattice correction [6,7]; Fang et al. provide the continuous conditional-entropy prediction argument [3]. Recent work applies the continuous argument to time-series complexity ranking, derives entropy-based lower bounds on the dimension of finite-time prediction spaces, and benchmarks finite-sample entropy estimators against certified entropy values [4,5,11]. The operational comparison appears once in main-text Table 1. The additional information-projection and recent estimator-benchmarking references above identify the specific inherited ingredients without repeating that table.

For reference, Table S21 separates the four questions answered by the DELB package.

Remark S1 (Claim hierarchy). *The contribution lies in jointly organizing the exact inverse-lattice floor, predictor-dependent refinement, necessary and sufficient attainability, and two-slack decomposition for non-anticipating integer prediction, and in connecting added side information to a reproducible finite-sample inferential chain. The discrete-Gaussian extremizer, KL identity, lattice correction, and Fano inequality remain inherited components.*

S8.2. Closed-Form, Temporal, and Vector Extensions

Corollary S1 (Closed-form DELB). *Every predictor in Theorem 1 satisfies*

$$D_t^{(m)} \geq \left[\frac{2^{2H(X_t|\mathcal{F}_{t-m})}}{2\pi e} - \frac{1}{12} \right]_+. \quad (3)$$

The $1/12$ term is the lattice correction produced by uniform dequantization. This explicit expression is conservative relative to the exact inverse-lattice DELB and is not generally attainable.

Proposition S1 (Forecast-horizon monotonicity). *If $1 \leq \ell < m$, then*

$$H(X_t | \mathcal{F}_{t-m}) \geq H(X_t | \mathcal{F}_{t-\ell}), \quad (4)$$

and therefore

$$\mathcal{H}_{\mathbb{Z}}^{-1}[H(X_t | \mathcal{F}_{t-m})] \geq \mathcal{H}_{\mathbb{Z}}^{-1}[H(X_t | \mathcal{F}_{t-\ell})]. \quad (5)$$

Corollary S2 (Stationary one-step limit). *For a strictly stationary integer process with $\mathcal{F}_{t-1} = \sigma(X_0, \dots, X_{t-1})$, define*

$$\bar{H}(X) = \lim_{t \rightarrow \infty} H(X_t | X_0, \dots, X_{t-1}). \quad (6)$$

Then

$$\liminf_{t \rightarrow \infty} \mathbb{E}[(X_t - \hat{X}_t)^2] \geq \mathcal{H}_{\mathbb{Z}}^{-1}[\bar{H}(X)]. \quad (7)$$

Corollary S3 (Integer-vector closed-form DELB). *For $\mathbf{X}_t, \hat{\mathbf{X}}_t \in \mathbb{Z}^n$,*

$$\frac{1}{n} \mathbb{E} \|\mathbf{X}_t - \hat{\mathbf{X}}_t\|_2^2 \geq \left[\frac{2^{(2/n)H(\mathbf{X}_t|\mathcal{F}_{t-m})}}{2\pi e} - \frac{1}{12} \right]_+. \quad (8)$$

The proofs use uniform dequantization, information-set nesting, and the multivariate Gaussian maximum-entropy inequality, respectively, and are given in Appendix S8.

S8.3. Fano Counterpart for Classification Loss

Let a fixed quantizer map the integer target to

$$Y_t = Q(X_t) \in \{1, \dots, K\}, \quad (9)$$

and let $\hat{Y}_t$ be measurable with respect to $\mathcal{F}_{t-m}$. Write $P_e = \Pr(\hat{Y}_t \neq Y_t)$, and define

$$\phi_K(p) = h_2(p) + p \log_2(K-1), \quad 0 \leq p \leq 1 - \frac{1}{K}. \quad (10)$$

Proposition S2 (Fano classification-error floor). *For every such classifier,*

$$H(Y_t | \mathcal{F}_{t-m}) \leq h_2(P_e) + P_e \log_2(K-1), \quad (11)$$

and hence

$$P_e \geq \phi_K^{-1}[H(Y_t | \mathcal{F}_{t-m})]. \quad (12)$$

For $K = 2$, the correct increasing branch gives

$$P_e \geq h_2^{-1}[H(Y_t | \mathcal{F}_{t-m})], \quad h_2^{-1} : [0, 1] \rightarrow [0, 1/2]. \quad (13)$$

For the baseline and augmented information sets defined in Section 4.7, define

$$P_F^0 = \phi_K^{-1}[H(Y | \mathcal{F}^0)], \quad P_F^B = \phi_K^{-1}[H(Y | \mathcal{F}^B)]. \quad (14)$$

Corollary S4 (Side-information ordering of Fano floors). *The entropy reduction*

$$H(Y | \mathcal{F}^0) - H(Y | \mathcal{F}^B) = I(Y; Q_B(\mathbf{B}^{(q)} | \mathcal{F}^0) \geq 0 \quad (15)$$

implies

$$P_F^B \leq P_F^0. \quad (16)$$

This is a classification-loss counterpart, not a second derivation of the count-MSE DELB. The targets, losses, equality conditions, and numerical scales remain distinct.

S8.4. Conditional Integer Translations

Lemma S1 (Entropy invariance under a measurable integer translation). *Let X be integer valued, let $\mathcal{F}$ be a sigma-algebra, and let $\hat{X}$ be an integer-valued, $\mathcal{F}$ -measurable random variable. If $E = X - \hat{X}$, then*

$$H(E | \mathcal{F}) = H(X | \mathcal{F}). \quad (17)$$

Consequently,

$$H(E) = H(X | \mathcal{F}) + I(E; \mathcal{F}). \quad (18)$$

Proof. Condition on a realization of the information in $\mathcal{F}$. The value $\hat{x}$ is then fixed, and

$$\Pr(E = k | \mathcal{F}) = \Pr(X = k + \hat{x} | \mathcal{F}).$$

The map $k \mapsto k + \hat{x}$ is a bijection of $\mathbb{Z}$, so it only relabels the atoms of the conditional probability mass function. Shannon entropy is invariant under such a relabeling. Averaging the pointwise identity over $\mathcal{F}$ proves Equation (17). Finally, $I(E; \mathcal{F}) = H(E) - H(E | \mathcal{F})$, which gives Equation (18). $\square$

S8.5. The Integer Maximum-Entropy Distribution

Lemma S2 (Discrete-Gaussian extremizer). *For every $D > 0$, the entropy envelope in Definition 1 is attained by the centered discrete Gaussian in Equation (8), with the unique parameter $\lambda > 0$ satisfying $D(\lambda) = D$. Its entropy is given by Equation (10). For $D = 0$, the extremizer is the point mass at zero.*

Proof. Write

$$\Theta(\lambda) = \sum_{k \in \mathbb{Z}} \exp(-\lambda k^2), \quad q_\lambda(k) = \frac{\exp(-\lambda k^2)}{\Theta(\lambda)}.$$

Termwise differentiation is valid for every $\lambda > 0$, and therefore

$$\mathbb{E}_{q_\lambda}[Z^2] = -\frac{d}{d\lambda} \ln \Theta(\lambda) = D(\lambda).$$

Moreover,

$$\frac{dD(\lambda)}{d\lambda} = -\text{Var}_{q_\lambda}(Z^2) < 0.$$

The limits are $D(\lambda) \rightarrow \infty$ as $\lambda \downarrow 0$ and $D(\lambda) \rightarrow 0$ as $\lambda \uparrow \infty$. Hence each $D > 0$ corresponds to a unique λ .

Let p be any probability mass function on $\mathbb{Z}$ satisfying $\mathbb{E}_p[Z^2] \leq D$, and choose λ so that $D(\lambda) = D$. Nonnegativity of relative entropy gives

$$\begin{aligned} 0 &\leq D_{\text{KL}}(p\|q_\lambda) = \sum_{k \in \mathbb{Z}} p(k) \log_2 \frac{p(k)}{q_\lambda(k)} \\ &= -H(p) + \frac{\lambda \mathbb{E}_p[Z^2] + \ln \Theta(\lambda)}{\ln 2}. \end{aligned} \quad (19)$$

It follows that

$$H(p) \leq \frac{\lambda D + \ln \Theta(\lambda)}{\ln 2} = H(q_\lambda). \quad (20)$$

Equality requires both $D_{\text{KL}}(p\|q_\lambda) = 0$ and $\mathbb{E}_p[Z^2] = D$, hence $p = q_\lambda$. Direct substitution into $-\sum_k q_\lambda(k) \log_2 q_\lambda(k)$ yields Equation (10). If $D = 0$, the constraint forces $Z = 0$ almost surely, so the entropy is zero. $\square$

S8.6. Proof of the DELB and Predictor-Dependent Refinement

Proof of Theorem 1. Apply Lemma S1 with $\mathcal{F} = \mathcal{F}_{t-m}$. Then

$$H(E_t) = H(X_t | \mathcal{F}_{t-m}) + I(E_t; \mathcal{F}_{t-m}). \quad (21)$$

Because $E_t \in \mathbb{Z}$ and $\mathbb{E}[E_t^2] = D_t^{(m)}$, the definition of the entropy envelope implies

$$H(E_t) \leq \mathcal{H}_{\mathbb{Z}}(D_t^{(m)}).$$

The generalized inverse is nondecreasing. Thus,

$$D_t^{(m)} \geq \mathcal{H}_{\mathbb{Z}}^{-1}(H(E_t)),$$

and Equation (11) follows from Equation (21). Since mutual information is nonnegative, applying the same monotone inverse proves Equation (12).

By Lemma S2, equality in the predictor-dependent inequality holds precisely when E_t is the centered discrete Gaussian having second moment $D_t^{(m)}$, with the degenerate point mass at zero included. Equality in the universal inequality also requires $I(E_t; \mathcal{F}_{t-m}) = 0$, i.e., independence of the prediction error and the available information. $\square$

S8.7. Proof of the Exact Attainability Decomposition

Proof of Proposition 1. The calculation below specializes the standard maximum-entropy information-projection identity to the moment-matched discrete Gaussian [2,13]. It is included here to make the prediction-specific equality conditions and the two resulting slack terms self-contained.

First suppose $D > 0$, and write the moment-matched discrete Gaussian as

$$q_D(k) = \frac{\exp(-\lambda_D k^2)}{\Theta(\lambda_D)}, \quad \mathbb{E}_{q_D}[Z^2] = D.$$

Since $q_D(k) > 0$ for every $k \in \mathbb{Z}$, direct expansion of relative entropy in bits gives

$$\begin{aligned} D_{\text{KL},2}(P_E\|q_D) &= \sum_{k \in \mathbb{Z}} P_E(k) \log_2 \frac{P_E(k)}{q_D(k)} \\ &= -H(E) + \frac{\lambda_D \mathbb{E}[E^2] + \ln \Theta(\lambda_D)}{\ln 2} \\ &= \mathcal{H}_{\mathbb{Z}}(D) - H(E). \end{aligned} \quad (22)$$

The last equality uses $\mathbb{E}[E^2] = D$ and Equation (10). Lemma S1 gives

$$H(E) = H(X_t | \mathcal{F}) + I(E; \mathcal{F}).$$

Substitution into Equation (22) proves Equation (15); adding $I(E; \mathcal{F})$ to both sides proves Equation (16).

If $D = 0$, then $E = 0$ almost surely. Hence $q_0 = P_E = \delta_0$, $H(E) = 0$, $I(E; \mathcal{F}) = 0$, and $H(X_t | \mathcal{F}) = 0$ by conditional translation. Both equations therefore reduce to $0 = 0$.

Relative entropy and mutual information are nonnegative. Thus the first slack vanishes precisely when $P_E = q_D$, and the universal slack vanishes precisely when $P_E = q_D$ and $I(E; \mathcal{F}) = 0$. For discrete random variables, zero mutual information is equivalent to independence. $\square$

S8.8. Fano Classification-Error Results

Proof of Proposition S2. Let $J = \mathbf{1}\{\hat{Y}_t \neq Y_t\}$. Because J is determined by $(Y_t, \hat{Y}_t)$,

$$\begin{aligned} H(Y_t | \hat{Y}_t) &= H(J, Y_t | \hat{Y}_t) \\ &= H(J | \hat{Y}_t) + H(Y_t | J, \hat{Y}_t). \end{aligned}$$

The first term is at most $h_2(P_e)$. If $J = 0$, the second term is zero; if $J = 1$, at most $K - 1$ labels remain possible. Therefore

$$H(Y_t | \hat{Y}_t) \leq h_2(P_e) + P_e \log_2(K - 1).$$

Since $\hat{Y}_t$ is measurable with respect to $\mathcal{F}_{t-m}$, conditioning on the full information set cannot increase entropy:

$$H(Y_t | \mathcal{F}_{t-m}) \leq H(Y_t | \hat{Y}_t).$$

This proves Equation (11).

The function ϕ_K is continuous and strictly increasing on $[0, 1 - 1/K]$, from 0 to $\log_2 K$. If $P_e \leq 1 - 1/K$, monotone inversion gives Equation (12); if $P_e > 1 - 1/K$, that equation holds trivially because the inverse takes values only in $[0, 1 - 1/K]$. When $K = 2$, $\log_2(K - 1) = 0$, so $\phi_2 = h_2$ on $[0, 1/2]$, proving Equation (13). $\square$

Proof of Corollary S4. By the definition of $\mathcal{F}_{c,i,t}^B$, the conditional mutual-information identity gives

$$H(Y_{c,i,t} | \mathcal{F}_{c,i,t}^0) - H(Y_{c,i,t} | \mathcal{F}_{c,i,t}^B) = I(Y_{c,i,t}; Q_B(\mathbf{B}_{c,i,t}^{(q)}) | \mathcal{F}_{c,i,t}^0) \geq 0.$$

The inverse ϕ_K^{-1} on its increasing branch is nondecreasing. Applying it to the two conditional entropies proves Equation (16). Proposition S2 applies separately to every classifier measurable with respect to either information set. $\square$

S8.9. Uniform Dequantization and the Closed-Form Bound

Proof of Corollary S1. Let $U \sim \text{Unif}[-1/2, 1/2]$ be independent of E_t , and put $Y = E_t + U$. On the interval $[k - 1/2, k + 1/2)$, the density of Y equals $\Pr(E_t = k)$. The intervals are disjoint and have unit length; hence

$$h(Y) = H(E_t). \quad (23)$$

Furthermore,

$$\text{Var}(Y) = \text{Var}(E_t) + \frac{1}{12} \leq \mathbb{E}[E_t^2] + \frac{1}{12} = D_t^{(m)} + \frac{1}{12}. \quad (24)$$

The Gaussian maximizes differential entropy at fixed variance, so

$$H(E_t) = h(Y) \leq \frac{1}{2} \log_2 \left[2\pi e \left(D_t^{(m)} + \frac{1}{12} \right) \right]. \quad (25)$$

Rearrangement yields

$$D_t^{(m)} \geq \frac{2^{2H(E_t)}}{2\pi e} - \frac{1}{12}.$$

Equation (21) gives both the predictor-dependent version and, after dropping the nonnegative mutual information term, the universal version. Finally $D_t^{(m)} \geq 0$; taking the positive part proves Equation (3). $\square$

S8.10. Temporal and Vector Consequences

Proof of Proposition S1. For $1 \leq \ell < m$, causal nesting gives $\mathcal{F}_{t-m} \subseteq \mathcal{F}_{t-\ell}$. The conditional mutual-information identity therefore gives

$$H(X_t | \mathcal{F}_{t-m}) - H(X_t | \mathcal{F}_{t-\ell}) = I(X_t; \mathcal{F}_{t-\ell} | \mathcal{F}_{t-m}) \geq 0,$$

which is Equation (4). Monotonicity of $\mathcal{H}_{\mathbb{Z}}^{-1}$ proves Equation (5). $\square$

Proof of Corollary S2. By stationarity,

$$H(X_t | X_0, \dots, X_{t-1}) = H(X_0 | X_{-t}, \dots, X_{-1}).$$

The right-hand side is nonincreasing in t and converges to the entropy rate $\bar{H}(X)$. Applying Equation (12) at each t , and using the continuity and monotonicity of the inverse entropy envelope established by the discrete-Gaussian parametrization in Lemma S2, gives Equation (7). $\square$

Proof of Corollary S3. Let $\mathbf{E}_t = \mathbf{X}_t - \hat{\mathbf{X}}_t$, and let $\mathbf{U}$ be independent and uniformly distributed on $[-1/2, 1/2)^n$. The unit cubes $\mathbf{k} + [-1/2, 1/2)^n$, $\mathbf{k} \in \mathbb{Z}^n$, are disjoint, so the multivariate analogue of Equation (23) is

$$h(\mathbf{E}_t + \mathbf{U}) = H(\mathbf{E}_t).$$

Let Σ_E be the covariance matrix of $\mathbf{E}_t$. The multivariate Gaussian maximum-entropy inequality and the arithmetic–geometric mean inequality for the eigenvalues give

$$\begin{aligned} H(\mathbf{E}_t) &\leq \frac{1}{2} \log_2 \left[(2\pi e)^n \det \left(\Sigma_E + \frac{1}{12} \mathbf{I}_n \right) \right] \\ &\leq \frac{n}{2} \log_2 \left[2\pi e \left(\frac{\text{tr}(\Sigma_E)}{n} + \frac{1}{12} \right) \right] \\ &\leq \frac{n}{2} \log_2 \left[2\pi e \left(\frac{\mathbb{E} \|\mathbf{E}_t\|_2^2}{n} + \frac{1}{12} \right) \right]. \end{aligned} \quad (26)$$

Conditional integer-vector translation is bijective, so

$$H(\mathbf{E}_t) \geq H(\mathbf{E}_t | \mathcal{F}_{t-m}) = H(\mathbf{X}_t | \mathcal{F}_{t-m}).$$

Substitution into Equation (26), rearrangement, and nonnegativity of the MSE prove Equation (8). The joint conditional entropy is essential: replacing it with a sum of marginal entropies would generally discard spatial dependence. $\square$

S8.11. Specialization to Crime Counts and Bicycle Mobility

The crime-count result is a direct substitution of $X_t = C_{c,i,t}$ and $\mathcal{F}_{t-m} = \mathcal{F}_{c,i,t}^0$ into Theorem 1. This proves Equations (34) and (35); Corollary S1 proves Equation (36). If the target is top-coded, deterministic data processing gives Equation (37). Because $\mathcal{H}_{\mathbb{Z}}^{-1}$ is nondecreasing, inserting the smaller top-coded entropy produces a no-larger, and therefore conservative, lower bound for the original-count MSE.

Proof of Proposition 2. By Equation (32),

$$\mathcal{F}_{c,i,t}^B = \mathcal{F}_{c,i,t}^0 \vee \sigma\{Q_B(\mathbf{B}_{c,i,t}^{(q)})\}.$$

The defining identity for conditional mutual information gives

$$I(C_{c,i,t}; Q_B(\mathbf{B}_{c,i,t}^{(q)}) | \mathcal{F}_{c,i,t}^0) = H(C_{c,i,t} | \mathcal{F}_{c,i,t}^0) - H(C_{c,i,t} | \mathcal{F}_{c,i,t}^B),$$

which is Equation (41) and is nonnegative. Applying the nondecreasing function $\mathcal{H}_{\mathbb{Z}}^{-1}$ proves Equation (42). From the definition in Equation (43),

$$\frac{N_d(C | \mathcal{F}^B)}{N_d(C | \mathcal{F}^0)} = 2^{2[H(C|\mathcal{F}^B) - H(C|\mathcal{F}^0)]} = 2^{-2\Delta H^B},$$

which proves Equation (44).

Finally, conditional on $\mathcal{F}_{c,i,t}^0$, $Q_B(\mathbf{B}_{c,i,t}^{(q)})$ is a deterministic function of $\mathbf{B}_{c,i,t}^{(q)}$. The conditional Markov chain

$$C_{c,i,t} \longrightarrow \mathbf{B}_{c,i,t}^{(q)} \longrightarrow Q_B(\mathbf{B}_{c,i,t}^{(q)})$$

therefore satisfies the conditional data-processing inequality, proving Equation (45). $\square$

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